



Course outline

Batch: 2023-2027

Course : Advanced Deep Learning Architectures
(Core Subject)

Course Instructor: Purva Thakare.

Term: VI

Credits: 5.0

No. of Sessions: 50

OVERVIEW

Course Description	The course Advanced Deep Learning Architecture offers an in-depth study of modern deep learning architectures, beginning with foundational principles such as backpropagation, generalization, universal approximation, and VC bounds. It explores advanced training strategies including weight initialization, normalization, regularization, and optimization techniques for stable and efficient learning. The course covers state-of-the-art vision and sequence models such as ResNet/DenseNet variants, EfficientNet, ConvNeXt, transformers, and long-context attention mechanisms. It also introduces graph neural networks, diffusion-based generative models, and emerging architectures like memory-augmented networks and neural theorem provers. Finally, students learn scalable deployment techniques including distributed training, pruning, knowledge distillation, and federated learning.
Course Learning Objectives	By the end of this course, students will be able to understand and apply the theoretical foundations of deep learning, including universal approximation, VC bounds, and generalization concepts. They will gain the ability to analyze and design advanced neural architectures such as ResNet, DenseNet, EfficientNet, ConvNeXt, Vision Transformers, and long-context transformer models. Students will also learn modern optimization, normalization, regularization, and scalable parallel training techniques for efficient model development. Additionally, they will be able to implement attention-based, recurrent, transformer, and graph neural network models for vision, language, and structured data applications. Finally, the course will enable students to explore emerging paradigms such as diffusion-based generative models, neural reasoning systems, pruning, knowledge distillation, and federated learning for real-world deployment.
Pedagogy	

Course Requirements	Basic knowledge of machine learning, linear algebra, probability, and proficiency in Python programming with familiarity in deep learning frameworks such as PyTorch or TensorFlow.
Group Project	Students will work collaboratively in groups to conceptualize and develop an innovative deep learning solution for a real-world problem. The project will involve selecting an appropriate dataset, designing a suitable model architecture, and conducting systematic training and evaluation. Teams will be encouraged to demonstrate creativity, critical thinking, and effective problem-solving throughout the development process. The project will conclude with a comprehensive report and a professional presentation highlighting methodology, results, and key insights.
Case Study Assignments	Case study assignments will immerse students in real-world deep learning challenges, enabling them to explore how advanced architectures are applied in practical systems. Students will investigate contemporary use cases, assess model design decisions, and derive meaningful insights from experimental results. These assignments foster analytical reasoning, innovation, and the ability to translate theoretical knowledge into impactful solutions. Through detailed reports and discussions, learners will strengthen their understanding of current trends and best practices in deep learning.

Basic Text Book	1. Ivan Vasilev,Advanced Deep Learning with Python:Design and Implement Advanced Next Generation AI-Solutions Using TensorFlow and PyTorch,Packt Publishing,2021/2022. 2. Ian Good fellow,Yoshua Bengio,and Aaron Courville,Deep-Learning,MITPress,November 2016. 3. Charu C.Aggarwal,Neural Networks and Deep-Learning:A Textbook,Springer,2023/2024.
Basic Reference Material	1. Witold Pedrycz and Shyi-Ming Chen (Eds.), Deep Learning: Concepts and Architectures, Springer Cham — A comprehensive overview of deep learning concepts, architectures and trends, useful as a reference for advanced topics. 2. Kolla Bhanu Prakash, et al. (Eds.), Advanced Deep Learning for Engineers and Scientists: APractical Approach, Springer — Offers practical insights and implementations of advanced deep learning methods suited for real applications. 3. M. Arif Wani, et al., Advances in Deep Learning, Springer — A focused reference on recent developments, deep network models, and advanced concepts in deep architectures. 4. GitHub— deep learning neural networks study guide — An open repository covering founda tional to advanced topics (CNNs, RNNs, Transformers, optimization, etc.) for deeper reading and example.
Reference Books	• Deep Learning by Ian Goodfellow, Yoshua Bengio & Aaron Courville • Neural Networks and Deep Learning by Michael A. Nielsen • Deep Learning with Python by François Chollet • Neural Networks and Deep Learning (Charu C. Aggarwal)

Class rules / Expectations from students	• Maintain regular attendance and actively participate in lectures, discussions, and lab sessions. • Complete assignments, case studies, and project tasks within the given deadlines with academic integrity. • Demonstrate consistent effort in learning both theoretical concepts and practical implementation skills. • Collaborate respectfully with peers during group activities and contribute meaningfully to team work. • Stay updated with recent developments in deep learning through self-study, research reading, and continuous practice.
Punctuality & Class Behavior	• Students are expected to be punctual and attend classes regularly, arriving on time and staying engaged throughout the session. • Maintain professional and respectful behavior, ensuring a focused and collaborative learning environment. • Active participation, discipline, and adherence to classroom norms are essential for effective learning.
Participation in Class Discussions	• Students are encouraged to actively participate in class discussions by asking questions, sharing insights, and engaging with course concepts. • Meaningful contributions and respectful interaction with peers will enhance collaborative learning and critical thinking.
E-Resource / Databases : (If Any)	• Awesome Deep Learning (GitHub) • arXiv.org • Analytics Vidya
Additional Reference Resources	
Videos : (If Any)	
AOL Evaluation / Any other Remarks	

TOPICS / SUB-TOPICS

Topic No.	1.0
Topic Name	Foundations of Deep Learning Architectures
Brief outline of deliverables	Module I will focus on developing a strong foundation in deep learning concepts through conceptual assignments and practical exercises on backpropagation, gradient stability, weight initialization, activation functions, normalization, and regularization techniques.
Book reference	
Pre-requisites for preparation	

Sub-Topic No.	Sub-Topic Name.	Brief outline of deliverables	Reference Book	Pre-requisites for preparation
1.1	Core Deep Learning Concepts: Neurons, Loss Functions, Backpropagation	Demonstrate understanding of neurons, loss functions, and backpropagation through conceptual and implementation-based learning outcomes.		
1.2	Universal Approximation, Bias–Variance Trade-off, and Generalization	Develop insights into approximation capability, bias–variance trade-off, and factors affecting model generalization.		
1.3	Vanishing/ Exploding Gradients: Causes and Mitigation	Identify vanishing/exploding gradient issues and apply mitigation strategies for stable deep network training.		
1.4	Weight Initialization: Xavier, He, Orthogonal	Implement and compare Xavier, He, and orthogonal initialization methods for improved convergence.		
1.5	Activation Functions: ReLU, Leaky ReLU, Swish, Mish, GELU	Evaluate the behavior and performance impact of modern activation functions in deep architectures.		

1.6	Normalization Techniques: BatchNorm, LayerNorm, RMSNorm	Apply BatchNorm, LayerNorm, and RMSNorm to enhance training stability and efficiency.		
1.7	Regularization Methods: Dropout Variants, Label Smoothing, Data Augmentation	Integrate dropout, label smoothing, and augmentation techniques to reduce overfitting and improve robustness.		

Topic No.	2.0
Topic Name	Advanced Sequence Models and Attention Architectures
Brief outline of deliverables	Module II will emphasize on understanding and implementation of efficient sequence and attention-based models through assignments and experiments on recurrent networks, multi-head and linear attention mechanisms, positional encodings, long-context transformers, and vision transformer architectures.
Book reference	
Pre-requisites for preparation	

Sub-Topic No.	Sub-Topic Name.	Brief outline of deliverables	Reference Book	Pre-requisites for preparation
2.1	Quasi-Recurrent and Advanced RNN Architectures	To understand and implement efficient recurrent network cells for sequence modeling.		
2.2	Bidirectional and Stacked Recurrent Networks	Deep and bidirectional recurrent designs to improve contextual learning in sequential data.		
2.3	Attention as Kernel Smoother & Core Attention Mechanisms	Conceptual clarity on attention intuition and its role in enhancing model focus and representation.		
2.4	Scaled Dot-Product and Multi-Head Attention	Multi-head attention mechanisms to capture diverse relational patterns in input sequences.		
2.5	Linear Attention Variants (Performer, Linformer, Nyströmformer)	Efficient attention alternatives designed for scalability on long sequences.		
2.6	Positional Encoding Techniques (Absolute, Relative, Rotary, ALiBi)	Positional encoding strategies to enable transformers to model sequence order effectively.		

2.7	Long-Context Transformer Extensions and Recurrence Models	Architectures that support extra-long context learning through recurrence and retention mechanisms.		
2.8	Vision Transformers and Efficient Transformer Variants	Competency in transformer-based vision architectures and their applications in image understanding.		

Topic No.	3.0
Topic Name	Convolutional Neural Networks and Computer Vision Architectures
Brief outline of deliverables	Module III involves practical exploration and implementation of modern vision architectures through assignments and projects on CNN evolution, ResNet/DenseNet/Inception models, efficient networks, Vision Transformers, and applications in classification, detection, and segmentation tasks.
Book reference	
Pre-requisites for preparation	

Sub-Topic No.	Sub-Topic Name.	Brief outline of deliverables	Reference Book	Pre-requisites for preparation
3.1	Evolution of CNNs to Deep Vision Systems	The progression from traditional CNNs to modern deep vision architectures and their design motivations.		
3.2	Residual Learning: ResNet and Variants	Residual connections for training deeper and more stable neural networks.		
3.3	Dense Connectivity: DenseNet Architectures	Dense feature reuse strategies for improved gradient flow and representation learning.		
3.4	Multi-Scale Feature Learning in the Inception Family	Implementation of Inception model for multi-scale feature extraction.		
3.5	Efficient CNN Architectures (MobileNet, ShuffleNet, EfficientNet)	Performance comparison of lightweight CNN architectures.		
3.6	Depthwise Separable Convolutions and Lightweight Model Design	Implementation showing reduced computation using depthwise separable convolution.		
3.7	Feature Pyramid Networks (FPN)	Implementation of multi-scale feature representation using FPN.		
3.8	Image Classification Application	CNN model for image classification.		

3.9	Object Detection using YOLO	YOLO-based real-time object detection.		
3.10	Image Segmentation using U-Net	U-Net model for image segmentation.		

Topic No.	4.0
Topic Name	Graph Neural Networks and Scalable Graph Learning
Brief outline of deliverables	Module IV focuses on Graph Neural Network architectures, scalable graph learning methods, dynamic graph modeling, and distributed graph processing techniques using modern frameworks
Book reference	
Pre-requisites for preparation	

Sub-Topic No.	Sub-Topic Name.	Brief outline of deliverables	Reference Book	Pre-requisites for preparation
4.1	Fundamentals of Graph Neural Networks	Introduction to graph representation learning and message passing mechanisms.		
4.2	Graph Convolutional Network (GCN)	Learning node representations using convolution operations on graph structures.		
4.3	GraphSAGE (Graph Sample and Aggregate)	Scalable node embedding generation through neighborhood sampling and aggregation.		
4.4	Graph Attention Network (GAT)	Use of attention mechanisms to assign importance to neighboring nodes.		
4.5	Graph Isomorphism Network (GIN)	Expressive graph representation learning with strong discriminative power.		
4.6	Spectral Graph Convolutions (Chebyshev Networks)	Spectral filtering methods for efficient graph convolution.		
4.7	Temporal and Dynamic Graph Modeling	Modeling and analysis of time-evolving graph structures.		

4.8	Oversmoothing and Oversquashing Issues	Understanding limitations in deep GNNs and their impact on information propagation.		
4.9	Large-Graph Sampling Techniques	Scalable training approaches for handling large graph datasets.		
4.10	Distributed Graph Learning using Deep Graph Library (DGL)	Implementation of scalable graph learning using the DGL framework		

Topic No.	5.0
Topic Name	Neural Reasoning, Model Compression and Federated Learning
Brief outline of deliverables	Implementation and evaluation of neural reasoning models, efficient compressed networks, and federated learning systems for scalable and privacy-preserving AI applications.
Book reference	
Pre-requisites for preparation	

Sub-Topic No.	Sub-Topic Name.	Brief outline of deliverables	Reference Book	Pre-requisites for preparation
5.1	Neural Theorem Provers and Neuro-Symbolic Reasoning Frameworks	Integration of neural networks with symbolic reasoning for logical inference.		
5.2	Logic Tensor Networks	Learning logical relationships using tensor-based neural architectures.		
5.3	Differentiable Neural Computers and Memory-Augmented Neural Networks	Neural models with external memory for complex reasoning and sequential tasks.		
5.4	Structured Pruning Techniques (Magnitude Pruning, Single-Shot Pruning, Movement Pruning)	Reducing model complexity by removing less important structured parameters.		
5.5	Unstructured Pruning and Lottery Ticket Hypothesis	Identification of sparse subnetworks capable of achieving comparable performance.		
5.6	Knowledge Distillation (Logit, Feature, and Task-Specific Distillation)	Knowledge Distillation (Logit, Feature, and Task-Specific Distillation)		
5.7	Federated Learning Paradigms (FedAvg, FedProx)	Distributed model training across decentralized data sources while preserving privacy.		

5.8	Applications in Scalable and Distributed AI Systems	Deployment of compressed and federated models for efficient large-scale AI systems.		
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EVALUATION PARAMETERS

Sr. No.	Parameter Name	Weightage	Exam Type	Exam Mode	Remarks
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